

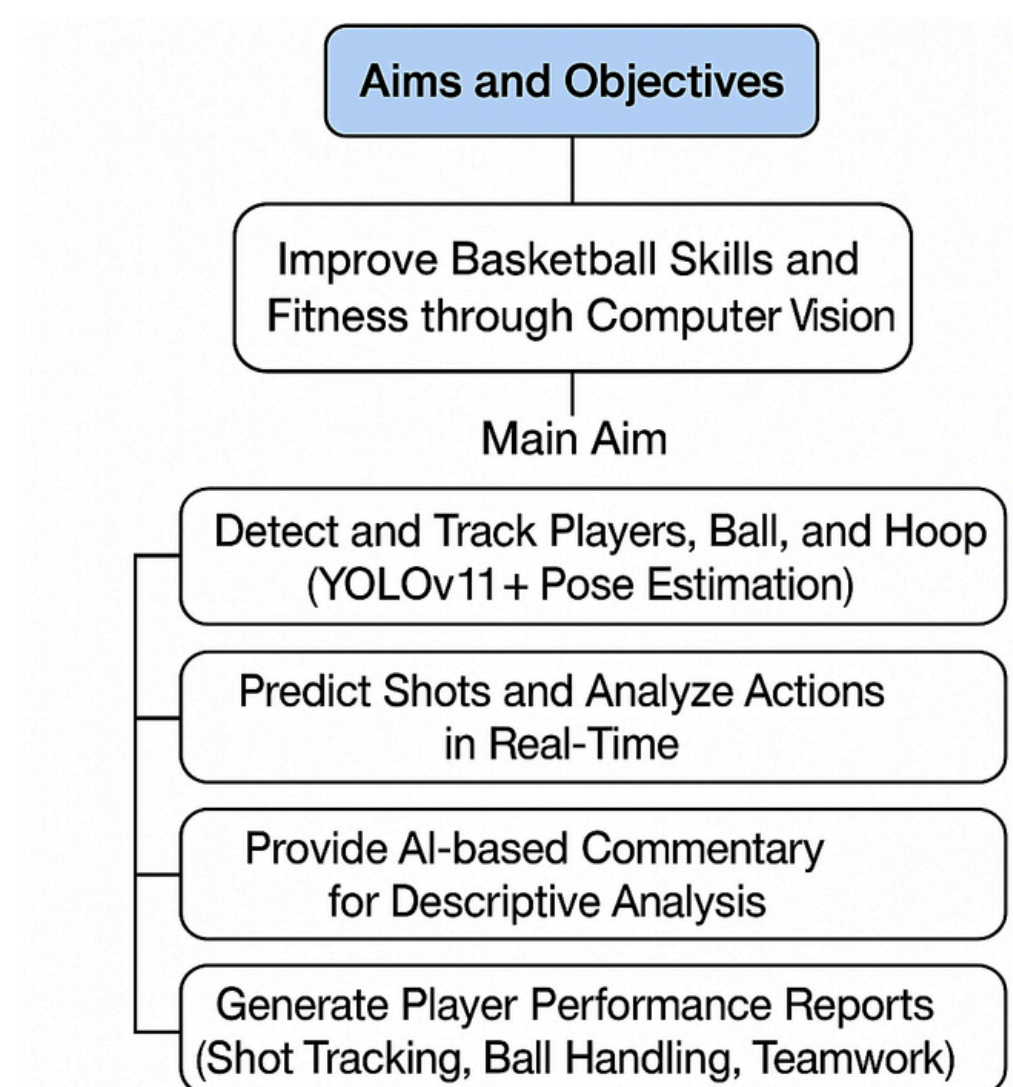


Enhancing Fitness and Basketball Skills Using Computer Vision

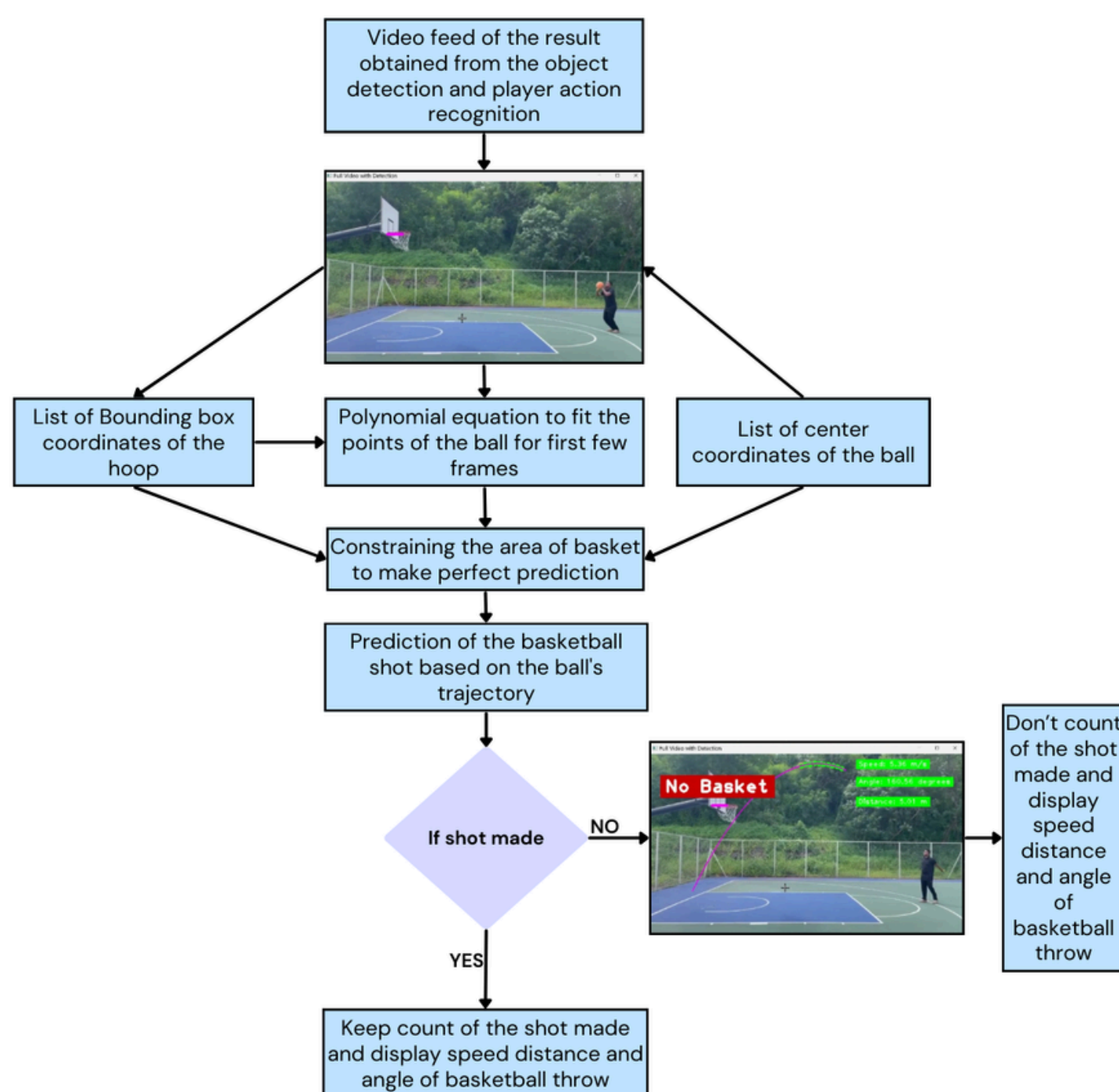
Introduction

- In modern basketball, understanding player movements, shot decisions, and gameplay dynamics is crucial for improving performance.
- Our project leverages YOLOv11-based detection to accurately identify players, the basketball, and the hoop in real-time.
- Combining this with Pose Estimation, we analyze player actions and predict shot outcomes with high precision.
- An added feature, AI-powered Commentary, enhances the system by providing real-time descriptive analysis, making it a complete tool for players, coaches, and analysts.

Aims and Objectives



Flow Diagram of Proposed Work



Methodology

- Data collection: Gather video footage of basketball games and practice sessions.
- Player detection and tracking: Use object detection algorithms (e.g., YOLO, Faster R-CNN).
- Action recognition: Implement CNN or LSTM-based models for action classification.
- Ball trajectory prediction: Track and predict ball movement using computer vision.
- Movement analysis: Utilize pose estimation 'OpenPose' for player movement tracking.
- Performance metrics calculation: Evaluate shot accuracy and dribble speed

Acknowledgements

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Teck Stack

- Programming Language: Python
- Libraries: OpenCV, TensorFlow / PyTorch
- Computer Vision Techniques: Object Detection (YOLOv11), Trajectory Prediction

Team Members

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